ECON 131 - INTERNATIONAL FINANCIAL MARKETS

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KEY TAKEAWAYS: LECTURE 7 (BONDS)

COMPOUNDING & NOMINAL VS. EFFECTIVE RATES

- **Compound Interest**: Calculated when interest is earned but not paid out:

$$1 + r = (1 + r_1)(1 + r_2)\cdots(1 + r_n)$$

- **Nominal Rate (i_{nom})**: Annualized w/o intra-year compounding.
- **Effective Annual Rate (r_{eff})**: True annual yield reflecting n compounding periods per year:

$$r_{\text{eff}} = \left(1 + \frac{i_{\text{nom}}}{n}\right)^n - 1$$

- **Present Value (PV) & Future Value (FV)**:

$$FV = PV(1 + r)^n \iff PV = \frac{FV}{(1 + r)^n}$$

BOND PRICING MECHANICS & YIELD TO MATURITY

- **Bond**: Asset promising periodic coupon payments plus principal (Face Value) at maturity.
- **General Valuation Equation**: Present value of coupon annuity plus principal repayment discounted at spot rates or yield:

$$PV = \sum_{t=1}^{n} \frac{\text{Coupon}_t}{(1+r_t)^t} + \frac{\text{Face Value}}{(1+r_n)^n}$$

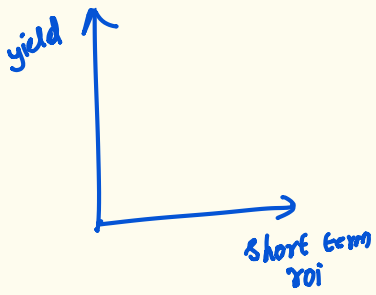
- When the discount rate r is constant across all periods:

$$PV = \text{Coupon} \times \left[\frac{1-(1+r)^{-n}}{r}\right] + \frac{\text{Face Value}}{(1+r)^n}$$

- **Yield to Maturity (YTM)**: The single internal rate of return (y) equating market price to the present value of all remaining cash flows.

$$\text{Market price} = \sum_{t=1}^{n} \frac{\text{Coupon}_t}{(1+ytm_t)^t} + \frac{F.V}{(1+ytm_t)^n}$$

(i) 3% coupon, 2 yr maturity.



$1000.

$30 → yr 1

$1000 + $30 → yr 2.

$$P = \$30 \times DF_1 + \$1030 \times DF_2.$$

(ii) 5% coupon

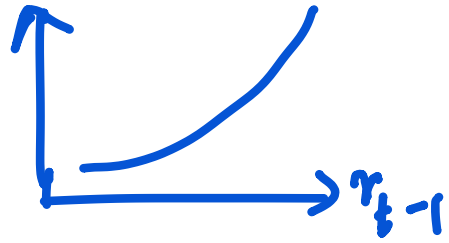
↳ $50 → yr 1 to 4

$1050 → yr 5

$$P = \$50 \cdot DF_1 + \$50 \cdot DF_2 + \dots + \$50 \cdot DF_4 + \$1050 \cdot DF_5$$

$$= \$50 \left[DF_1 + DF_2 + DF_3 + DF_4\right] + \$1050 \cdot DF_5$$

THE TERM STRUCTURE OF INTEREST RATES



- **Term Structure**: The relationship between spot interest rates (r_t) and time-to-maturity (t) for zero-coupon default-free bonds.
- **Empirical Regularity**: Term structure is upward sloping ($r_t > r_{t-1}$).
- **Three Theoretical Drivers**:

→ inflation risk.
→ interest risk

1. **Expectations Hypothesis**: Long-term yields are a geometric average of current short-term rates and market expectations of future short-term rates.
2. **Inflation Risk Premium**: Long-dated bonds face greater purchasing-power erosion risk (Fisher Equation: $1 + r_{\text{nom}} = [1 + r_{\text{real}}][1 + \pi]$).
3. **Interest Rate Risk**: Longer-duration bonds experience greater price volatility from unexpected shifts in market yields.

PS3 PROMPT & GUIDED PRACTICE

HOMEWORK PROBLEM: SPOT RATES & DCF VALUATION

$$DF_t = \frac{1}{(1+r_t)^t}$$

- **Given Term Structure**:

Year (t)	1	2	3	4	5
Spot Rate (r_t)	3.5%	4.2%	5.0%	6.5%	7.2%

$DF_1 = \frac{1}{(1+0.035)^1}$

$DF_2 = \frac{1}{(1+0.042)^2}$

DF_2 DF_3 DF_4 DF_5

- **Questions**:

1. What are the discount factors (DF_t) for each date $t = 1, \ldots, 5$?
2. Calculate the prices of the following Treasury bonds with \$1,000 par value using the DCF method (assume annual compounding):

F.V = \$1000

- ☐ 3% coupon, 2-year maturity
- ☐ 6% coupon, 5-year maturity
- ☐ 11% coupon, 5-year maturity

PARALLEL EXAMPLE: SETUP & DISCOUNT FACTORS

- **Parallel Term Structure (Changed Parameters)**:

$$r_1 = 3.0\%, \quad r_2 = 3.8\%, \quad r_3 = 4.5\%, \quad r_4 = 5.5\%, \quad r_5 = 6.2\%$$

$r_5 > r_4 > r_3 > r_2 > r_1$

- **Discount Factor Formula**: $DF_t = \frac{1}{(1+r_t)^t}$ $DF_5 < DF_4 < DF_3 < DF_2 < DF_1$
  - **Year 1**: $DF_1 = \frac{1}{(1.030)^1} = 0.970874$
  - **Year 2**: $DF_2 = \frac{1}{(1.038)^2} = 0.928122$
  - **Year 3**: $DF_3 = \frac{1}{(1.045)^3} = 0.876297$
  - **Year 4**: $DF_4 = \frac{1}{(1.055)^4} = 0.807217$
  - **Year 5**: $DF_5 = \frac{1}{(1.062)^5} = 0.740248$



- **Valuation Rule**: Price $P = \sum_{t=1}^{n} \text{Cash Flow}_t \times DF_t$

PARALLEL EXAMPLE: DCF BOND VALUATION

- **Bond 1 (4% Coupon, 2-Year Maturity, Par $1,000)**:
  - Cash flows: $C_1 = \$40$, $C_2 = \$1,040$
  - $P = (\$40 \times 0.970874) + (\$1,040 \times 0.928122) = \$1,004.08$
- **Bond 2 (5% Coupon, 5-Year Maturity, Par $1,000)**:
  - Cash flows: $C_{1\ldots4} = \$50/\text{yr}$, $C_5 = \$1,050$
  - $\sum_{t=1}^{4} DF_t = 0.970874 + 0.928122 + 0.876297 + 0.807217 = 3.582510$
  - $P = (\$50 \times 3.582510) + (\$1,050 \times 0.740248) = \$956.39$
- **Bond 3 (8% Coupon, 5-Year Maturity, Par $1,000)**:
  - Cash flows: $C_{1\ldots4} = \$80/\text{yr}$, $C_5 = \$1,080$
  - $P = (\$80 \times 3.582510) + (\$1,080 \times 0.740248) = \$1,086.07$